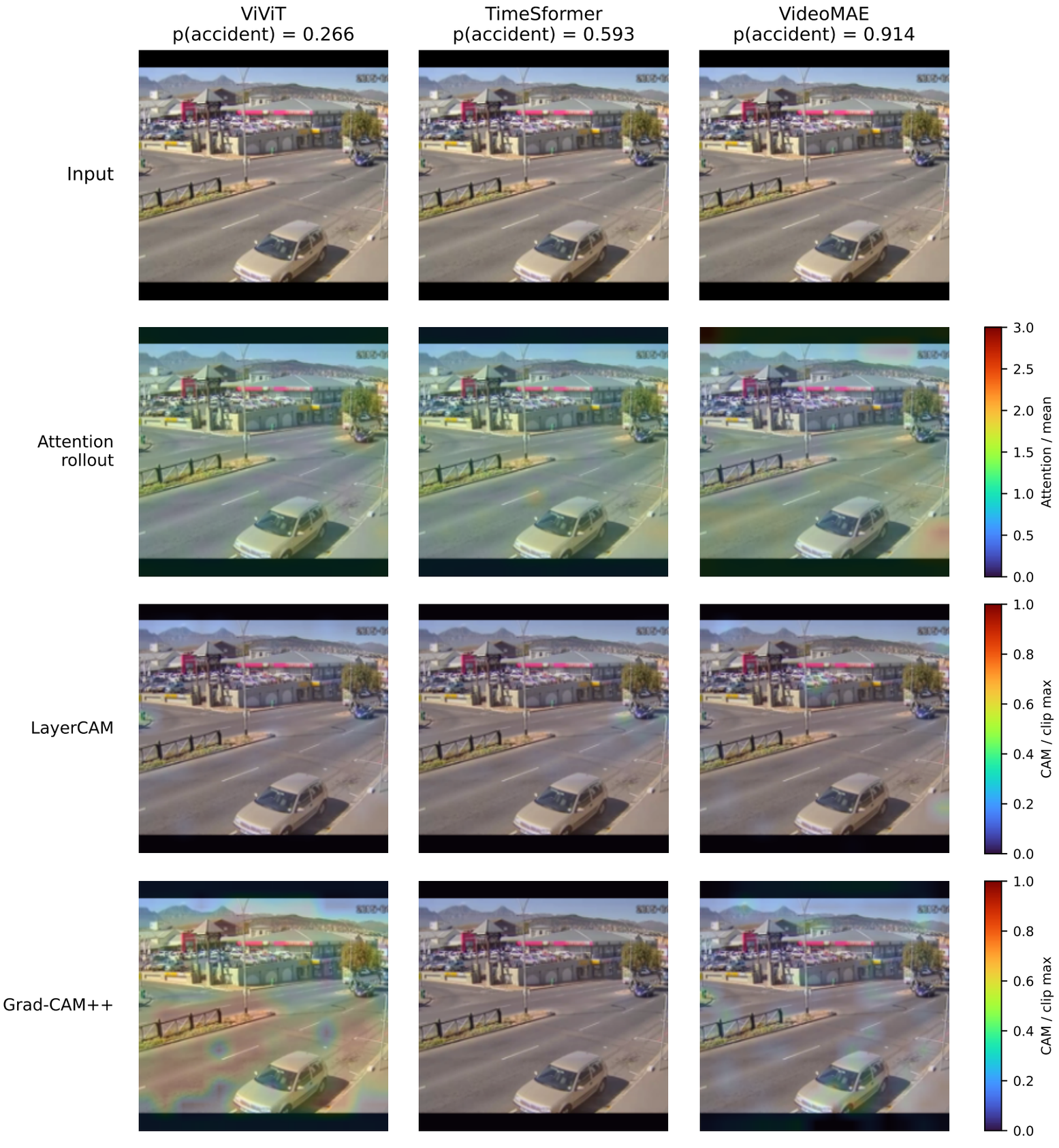
N01 | Non-accident (dataset label) | t = 8.14 s

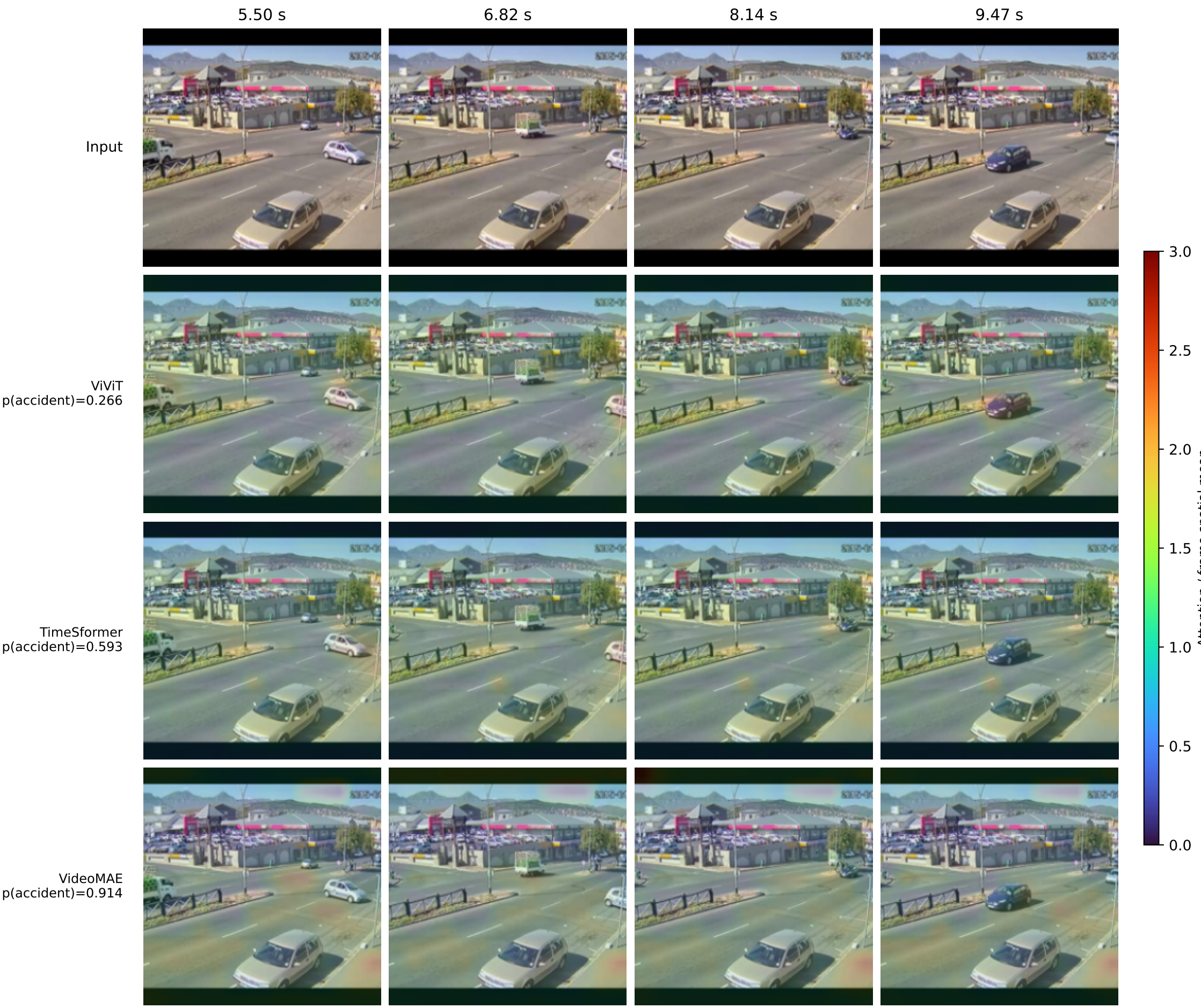
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N01 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N01 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

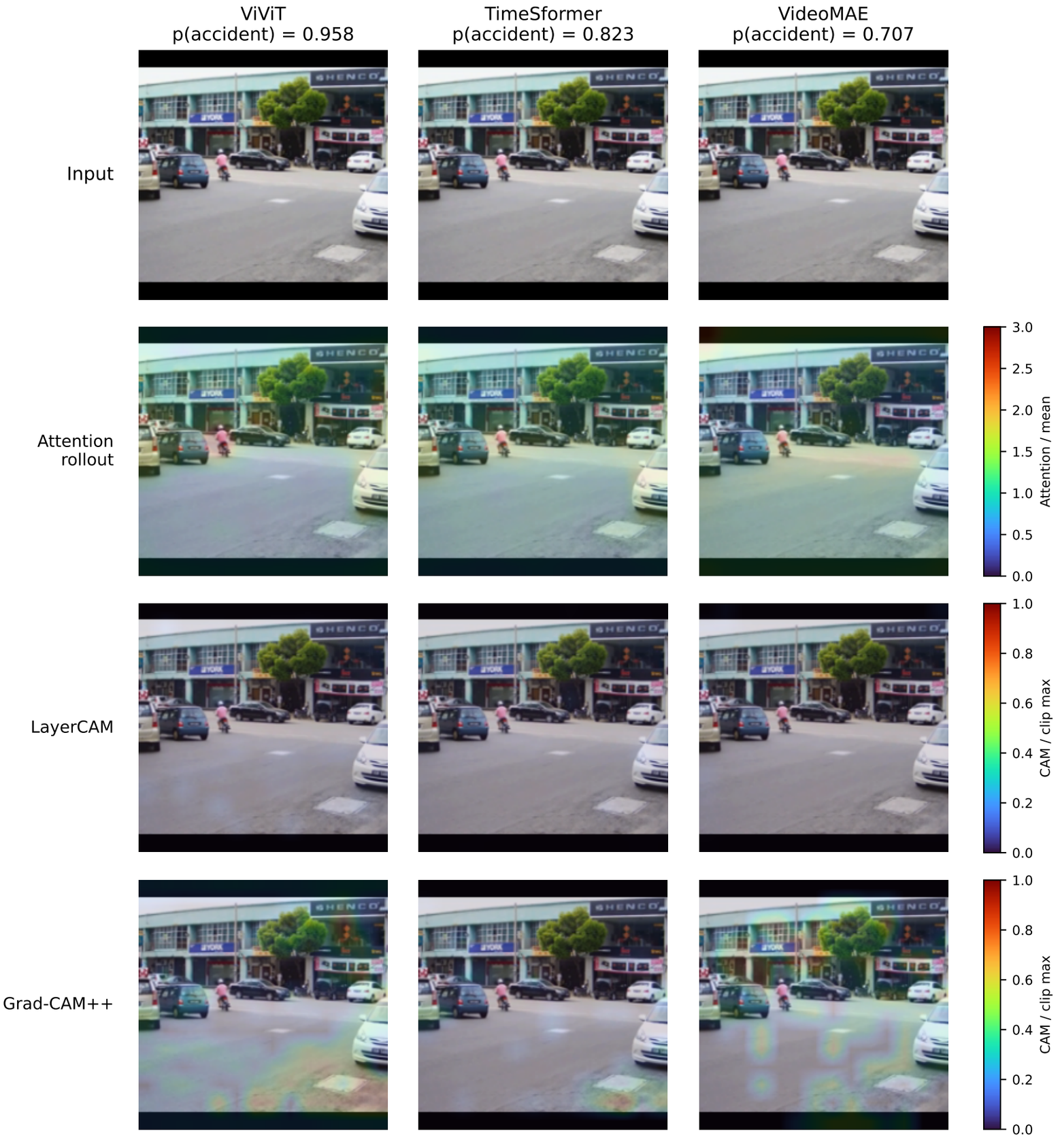
N01 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N02 | Non-accident (dataset label) | t = 22.61 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N02 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

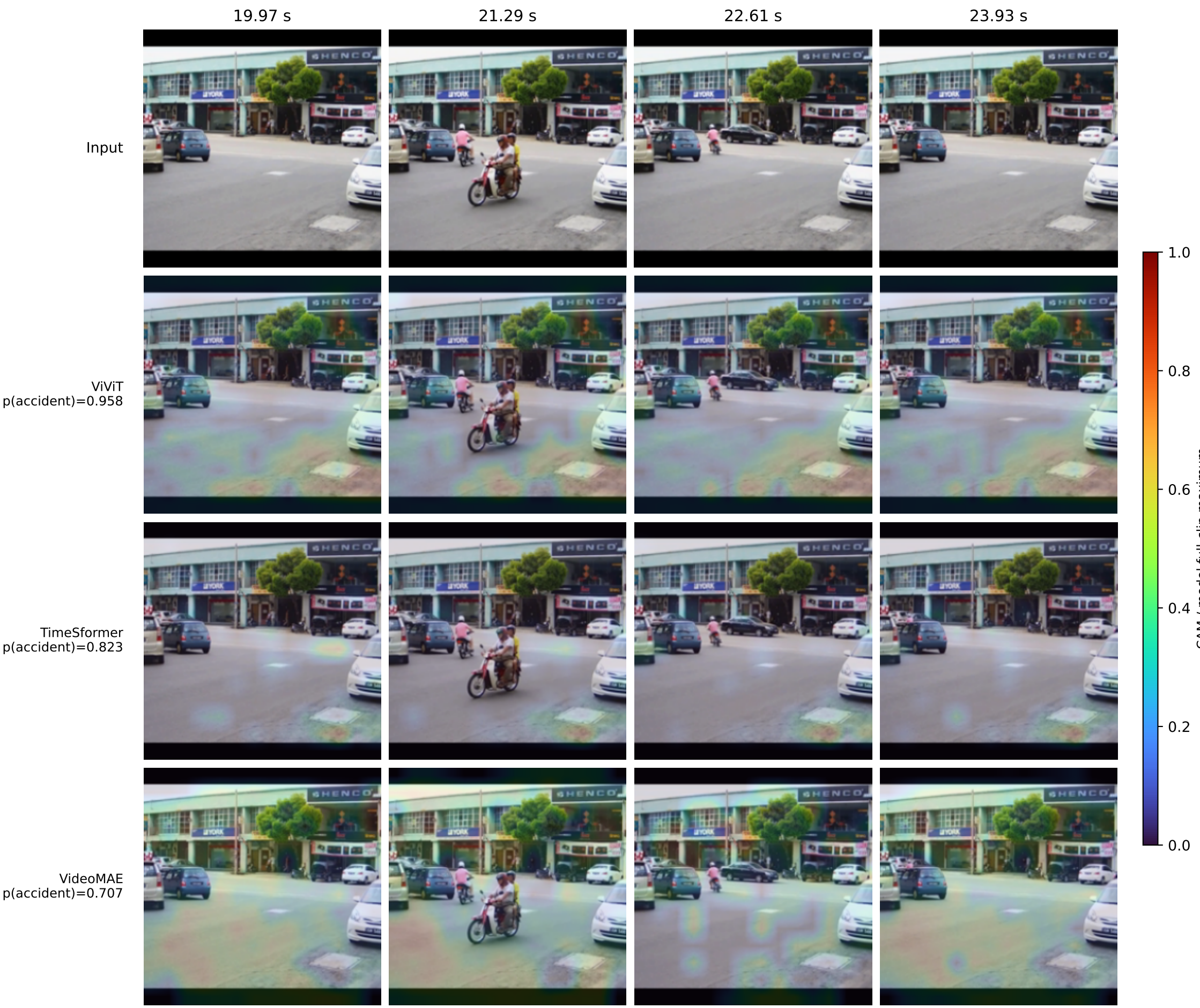
N02 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N02 | Grad-CAM++ | Non-accident label

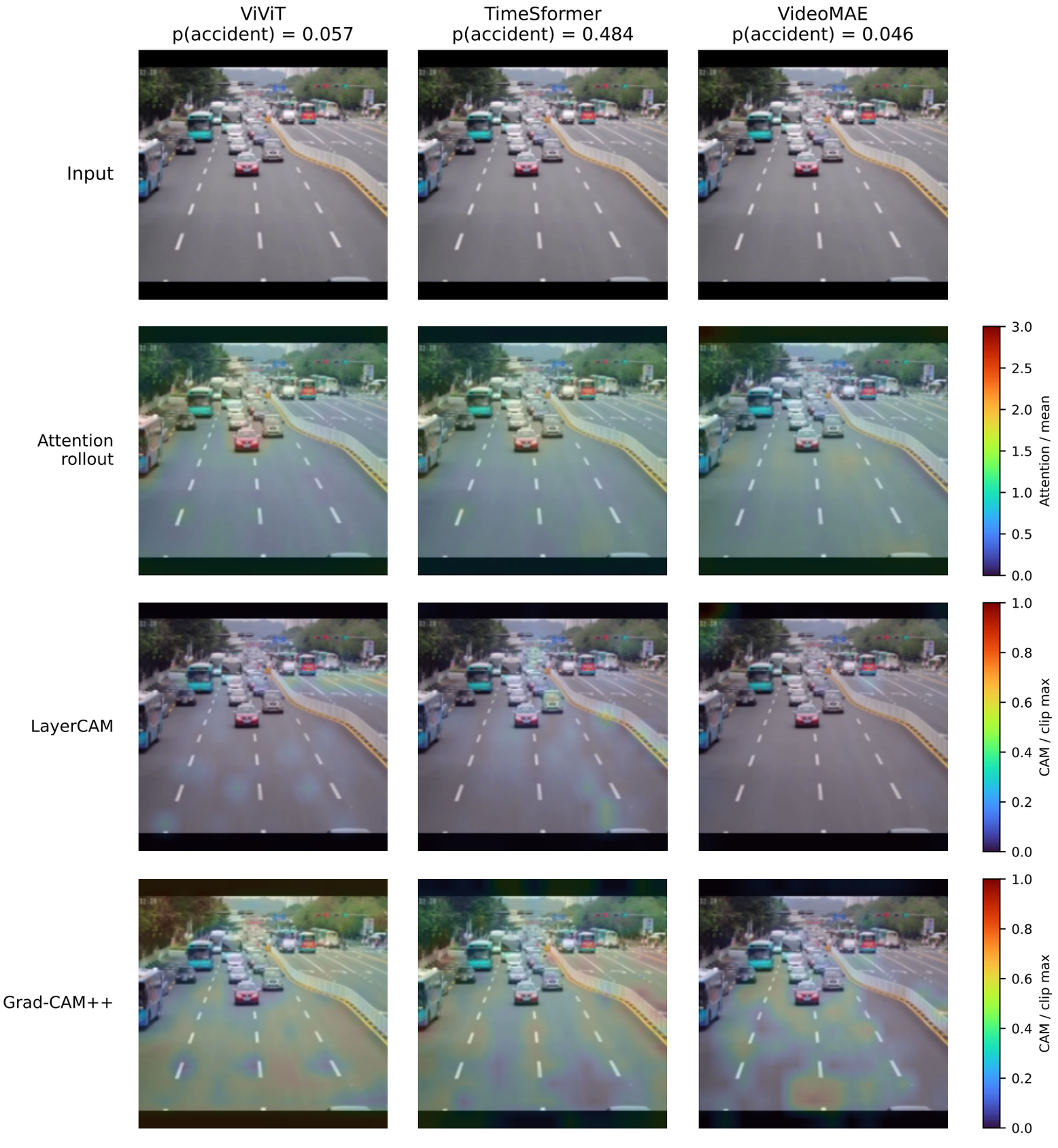
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N03 | Non-accident (dataset label) | t = 70.61 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N03 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

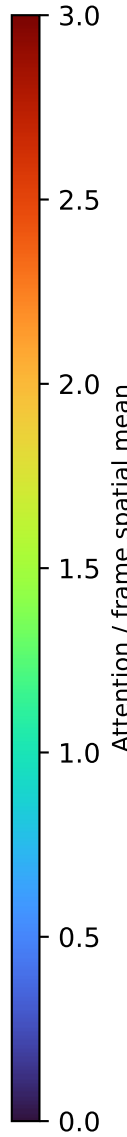
67.97 s | 69.29 s | 70.61 s | 71.93 s

Input

ViViT
p(accident)=0.057

TimeSformer
p(accident)=0.484

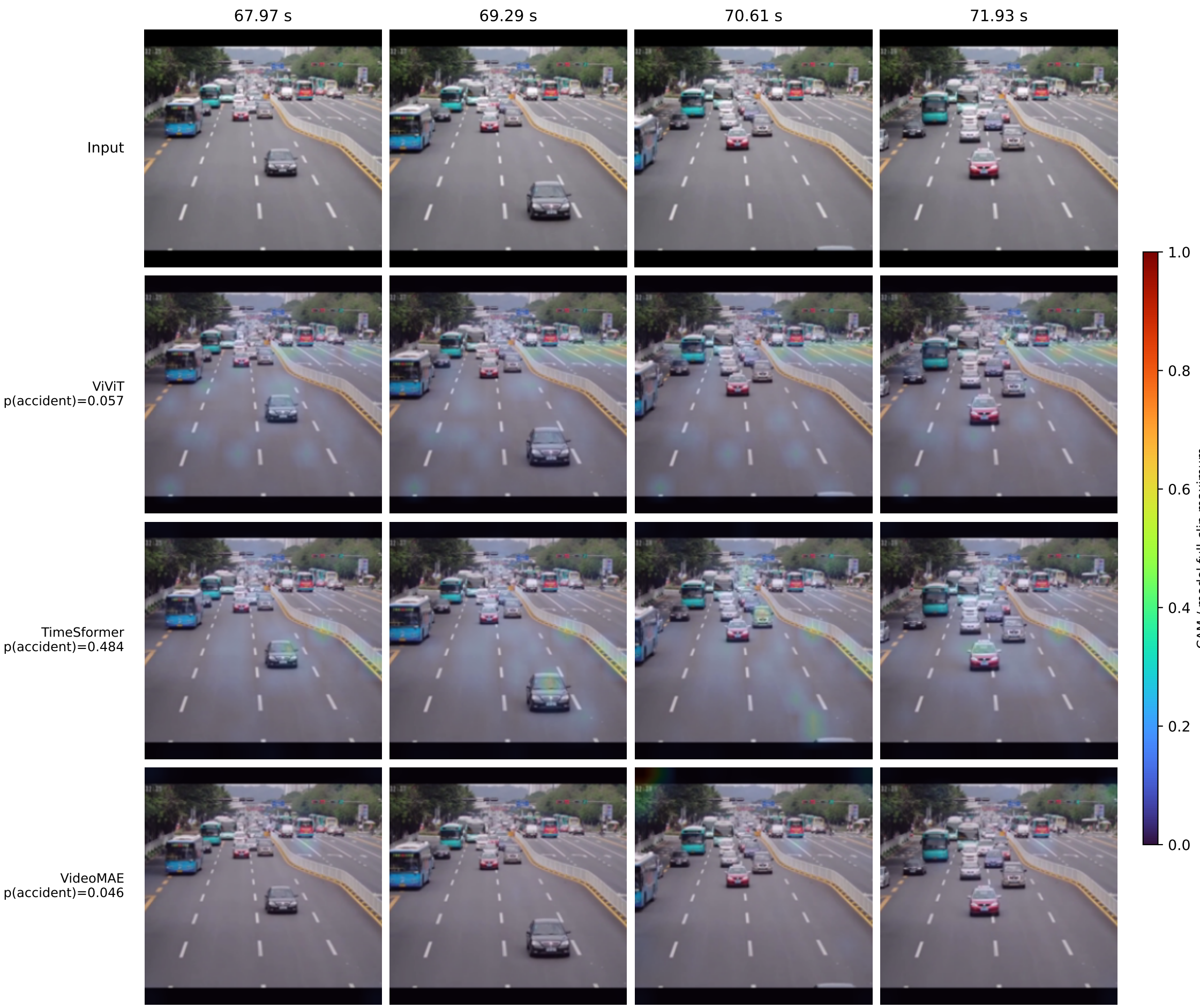
VideoMAE
p(accident)=0.046



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N03 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

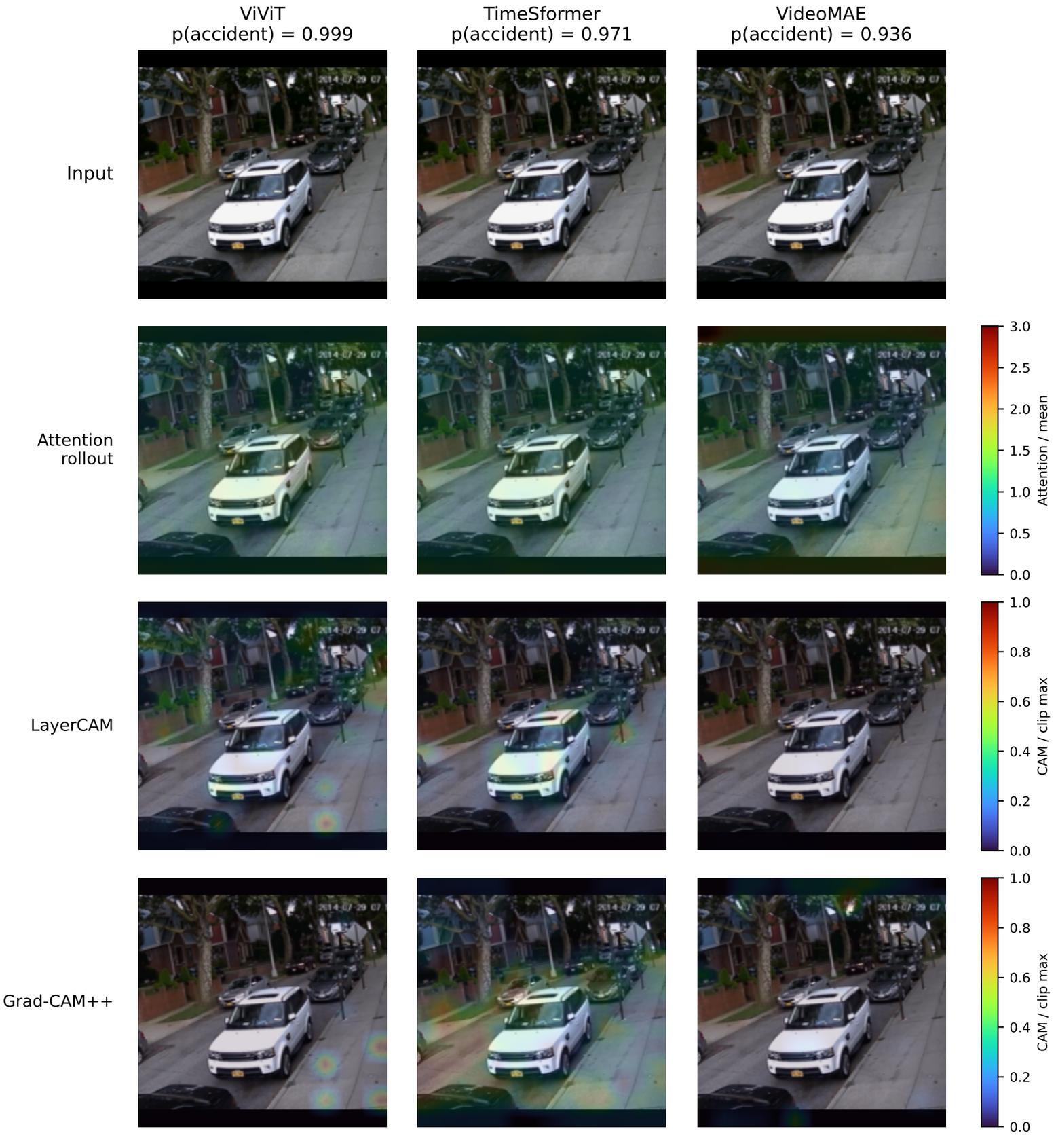
N03 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N04 | Non-accident (dataset label) | t = 39.94 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N04 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N04 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

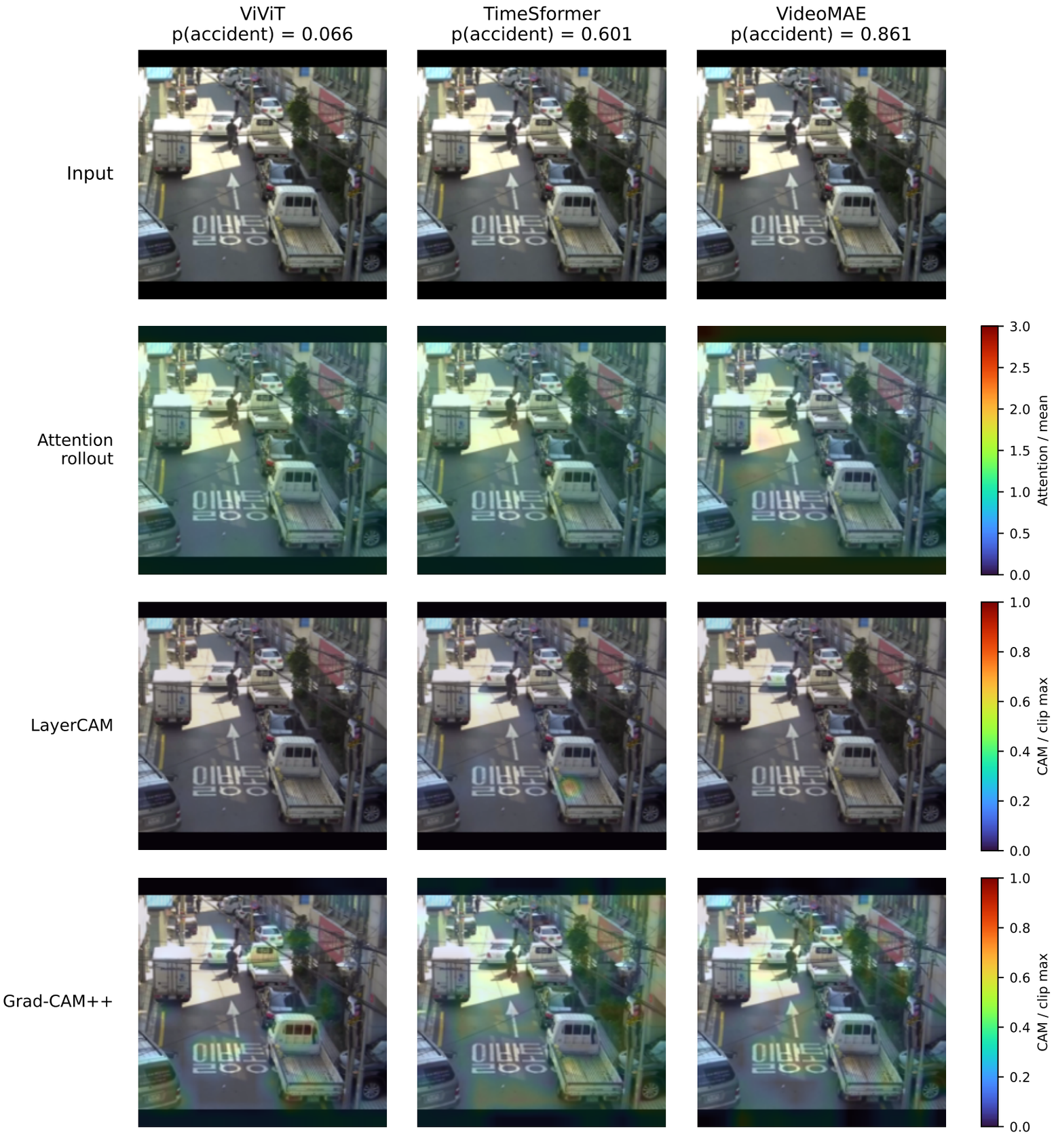
N04 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N05 | Non-accident (dataset label) | t = 29.64 s

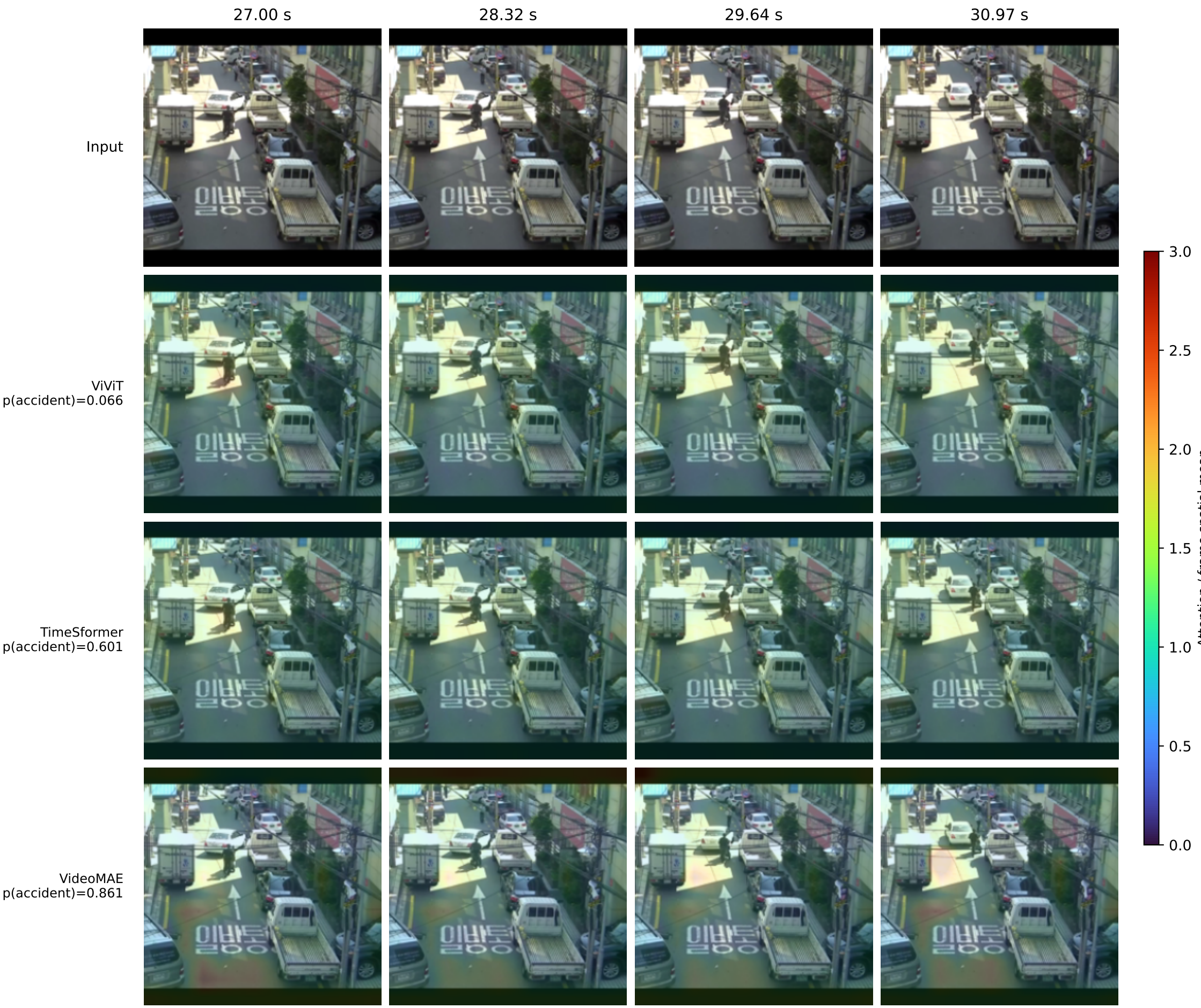
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N05 | Attention rollout | Non-accident label

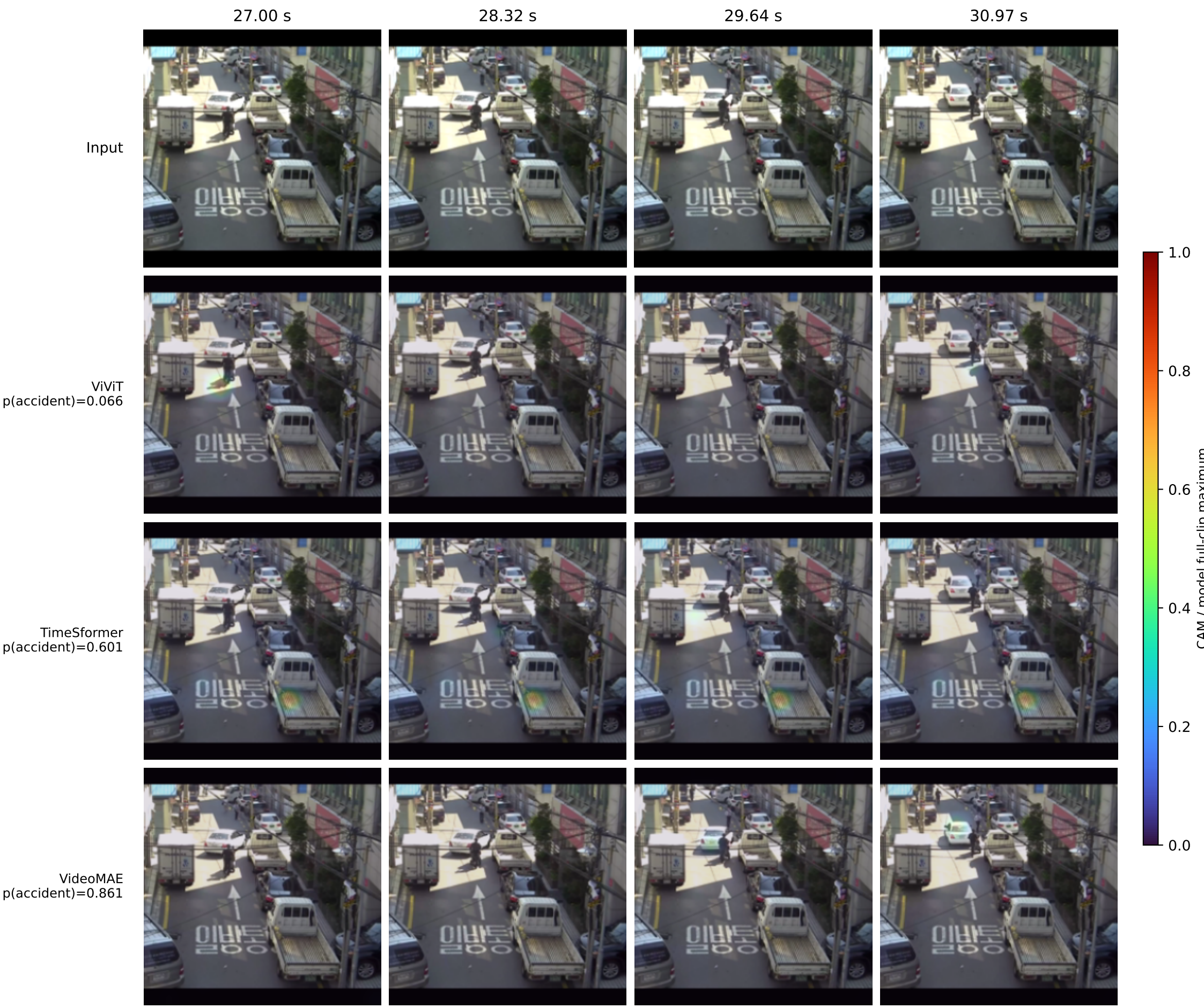
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N05 | LayerCAM | Non-accident label

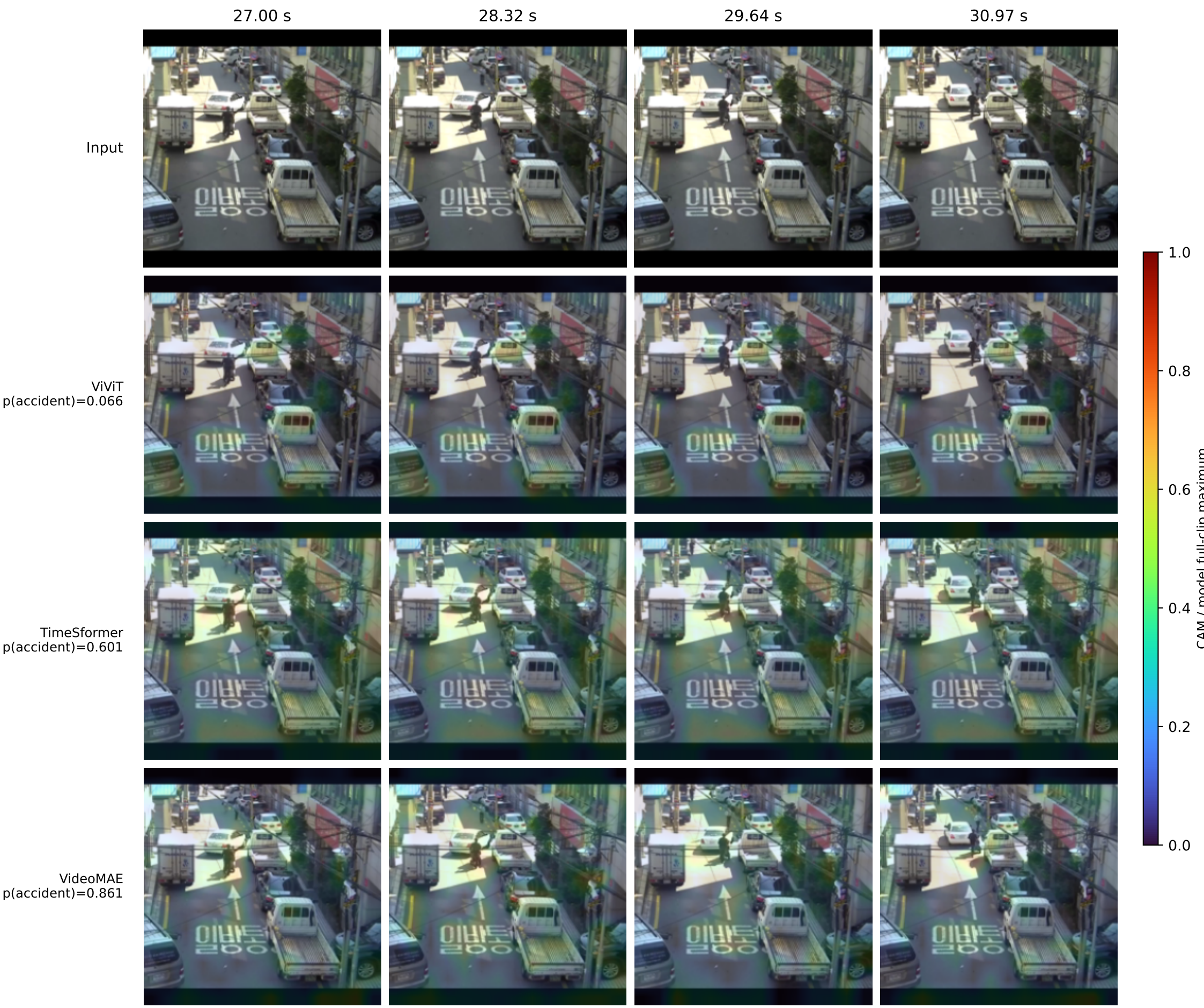
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N05 | Grad-CAM++ | Non-accident label

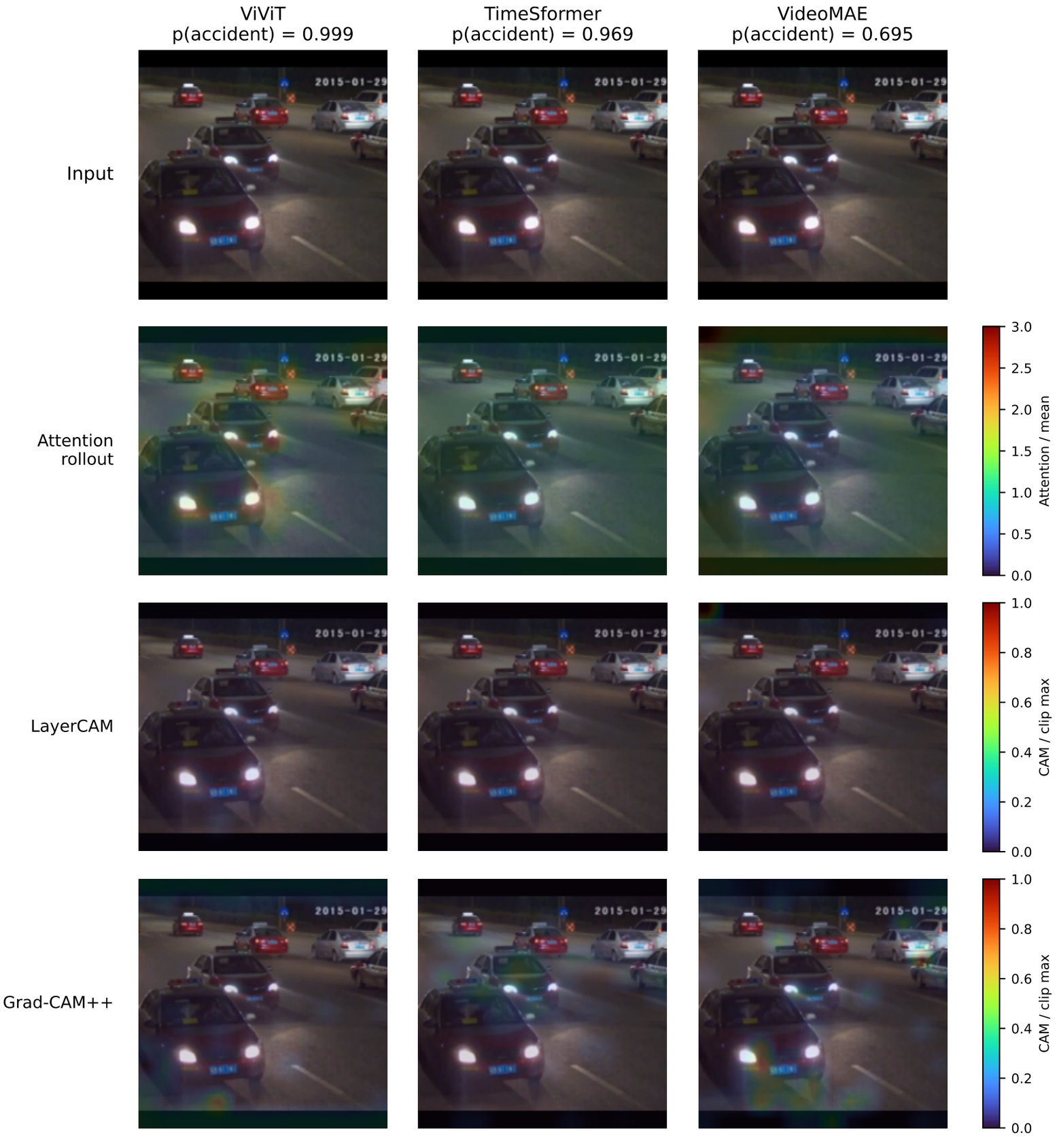
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N06 | Non-accident (dataset label) | t = 12.93 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N06 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N06 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

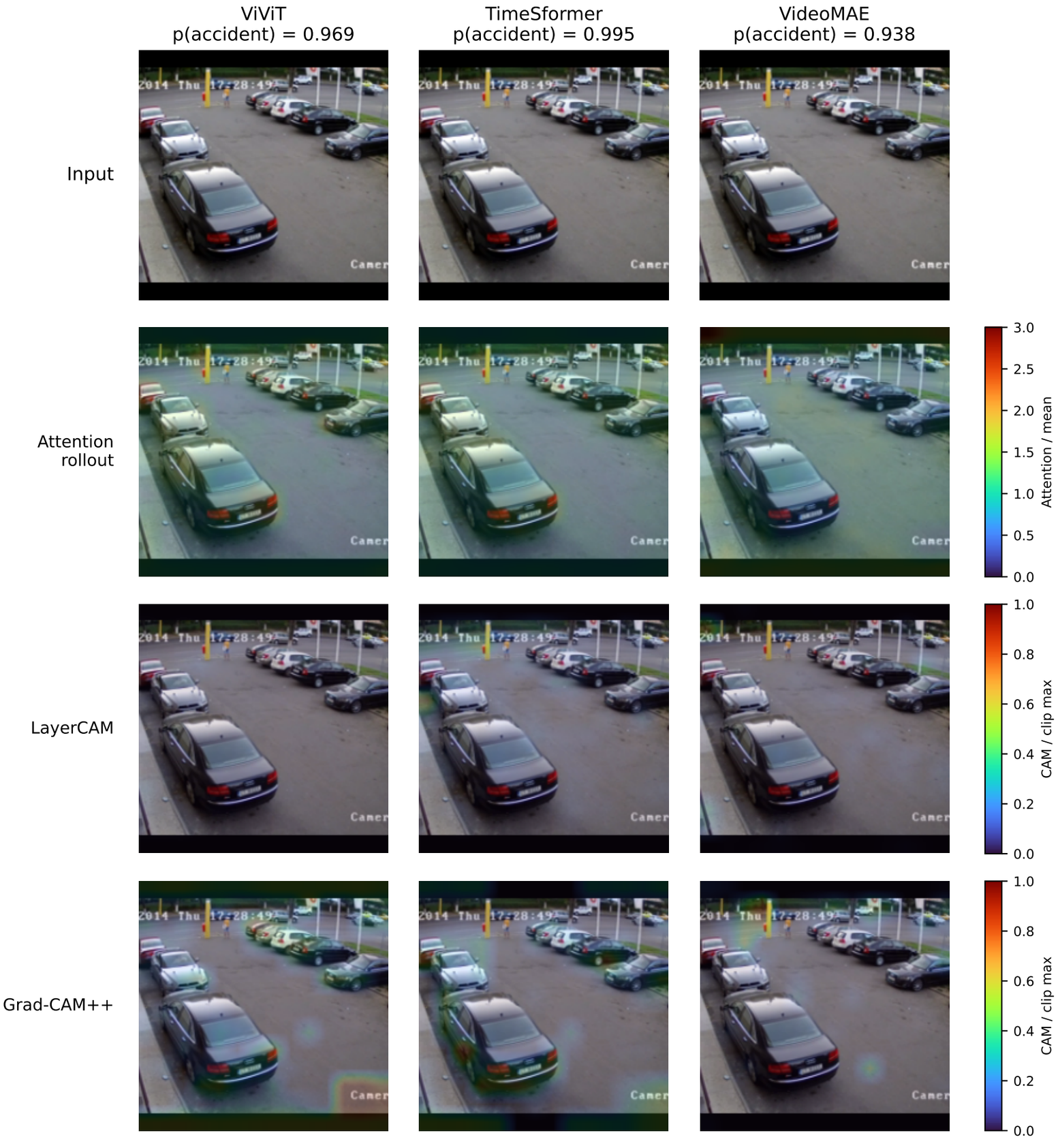
N06 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N07 | Non-accident (dataset label) | t = 43.49 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N07 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

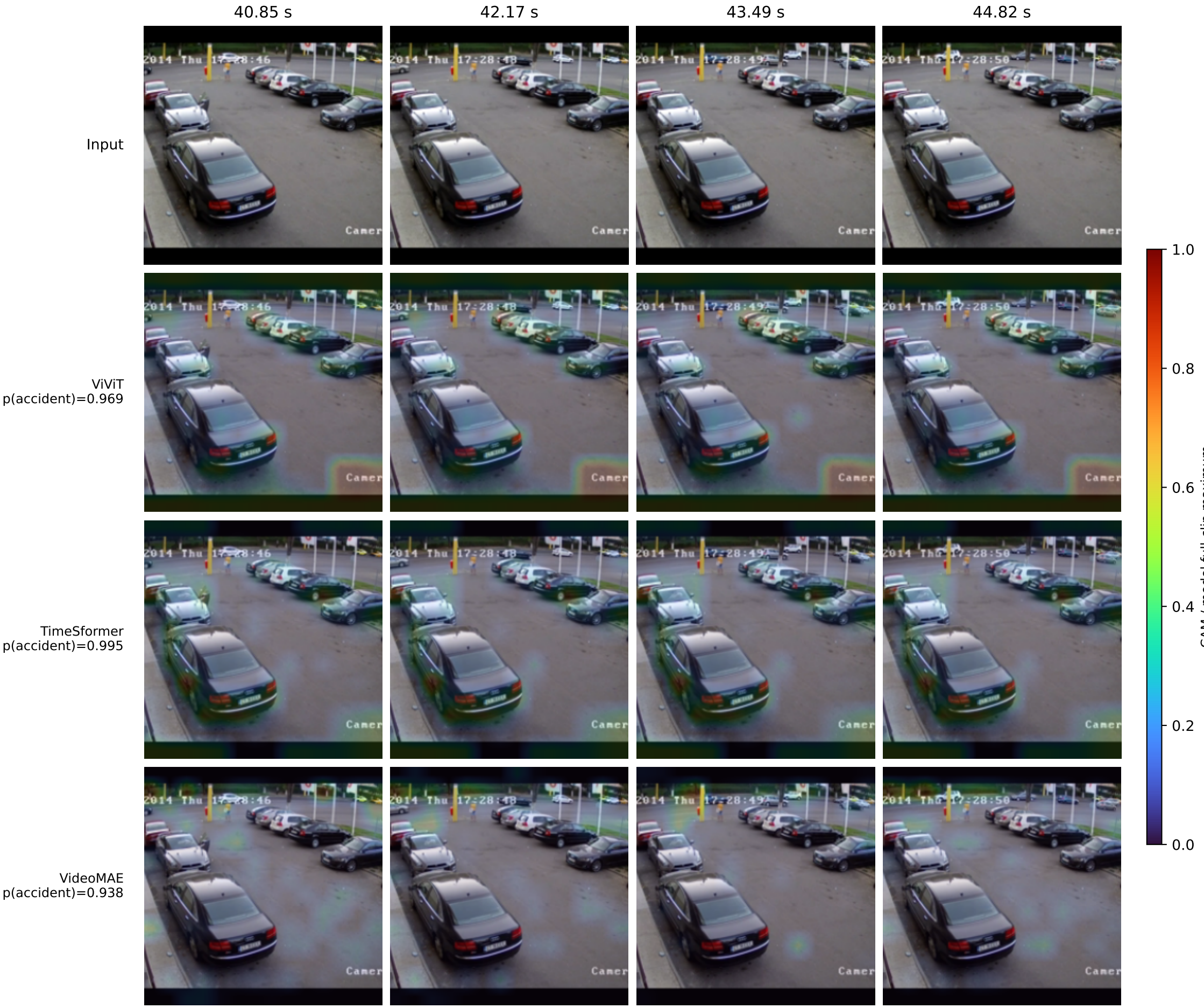
N07 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N07 | Grad-CAM++ | Non-accident label

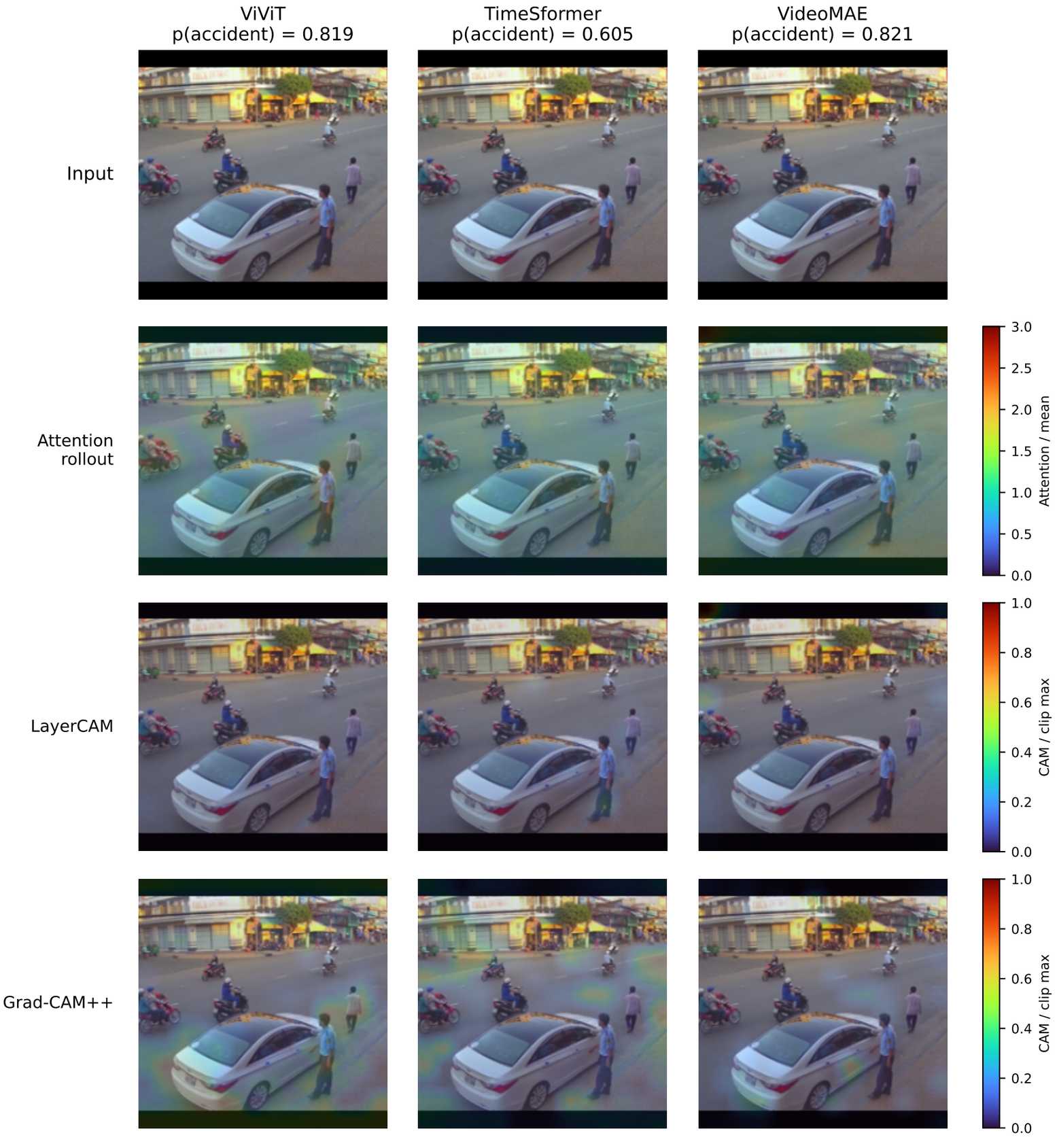
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N08 | Non-accident (dataset label) | t = 90.78 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N08 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N08 | LayerCAM | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

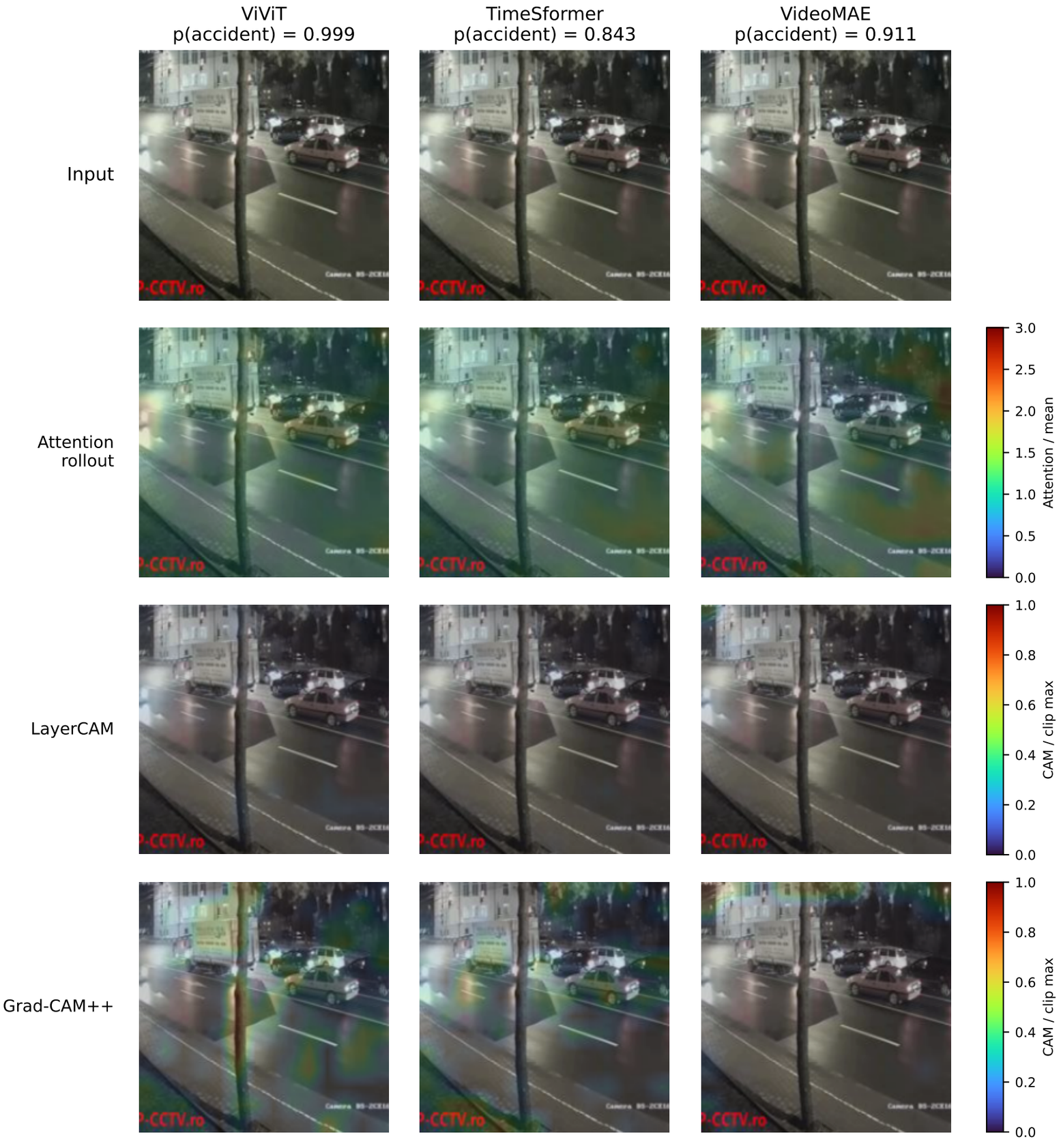
N08 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N09 | Non-accident (dataset label) | t = 75.68 s

Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

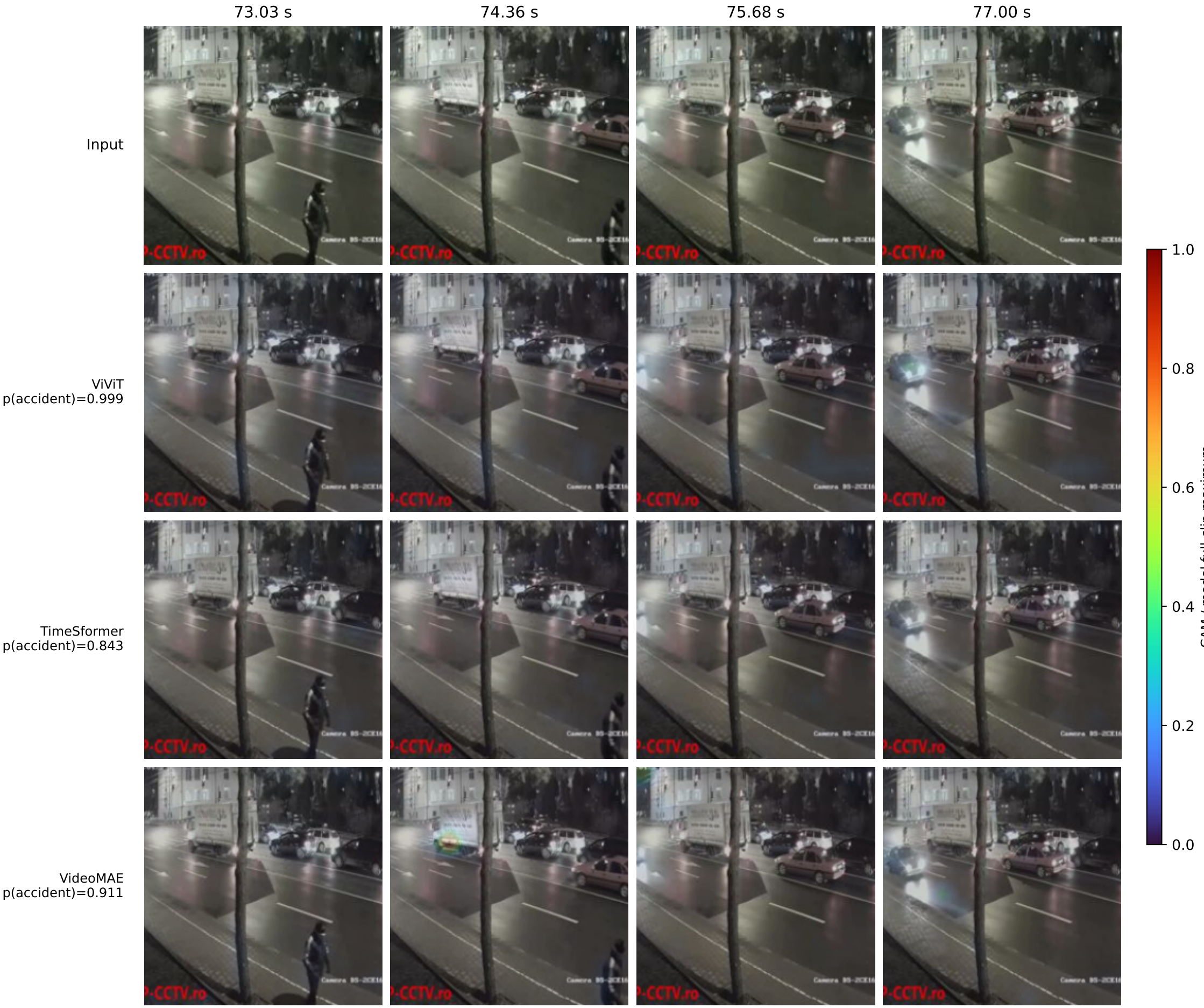
N09 | Attention rollout | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42

Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N09 | LayerCAM | Non-accident label

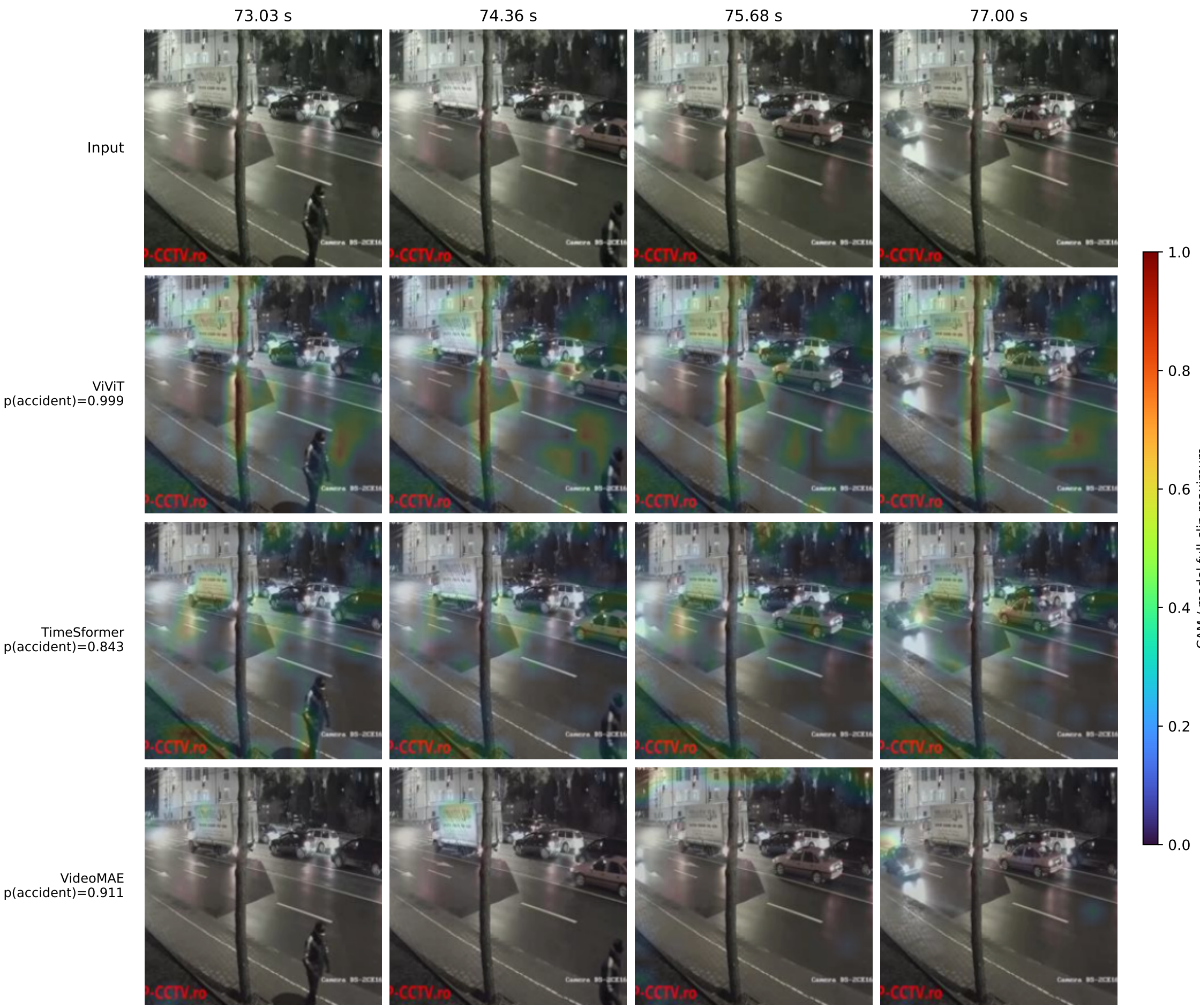
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N09 | Grad-CAM++ | Non-accident label

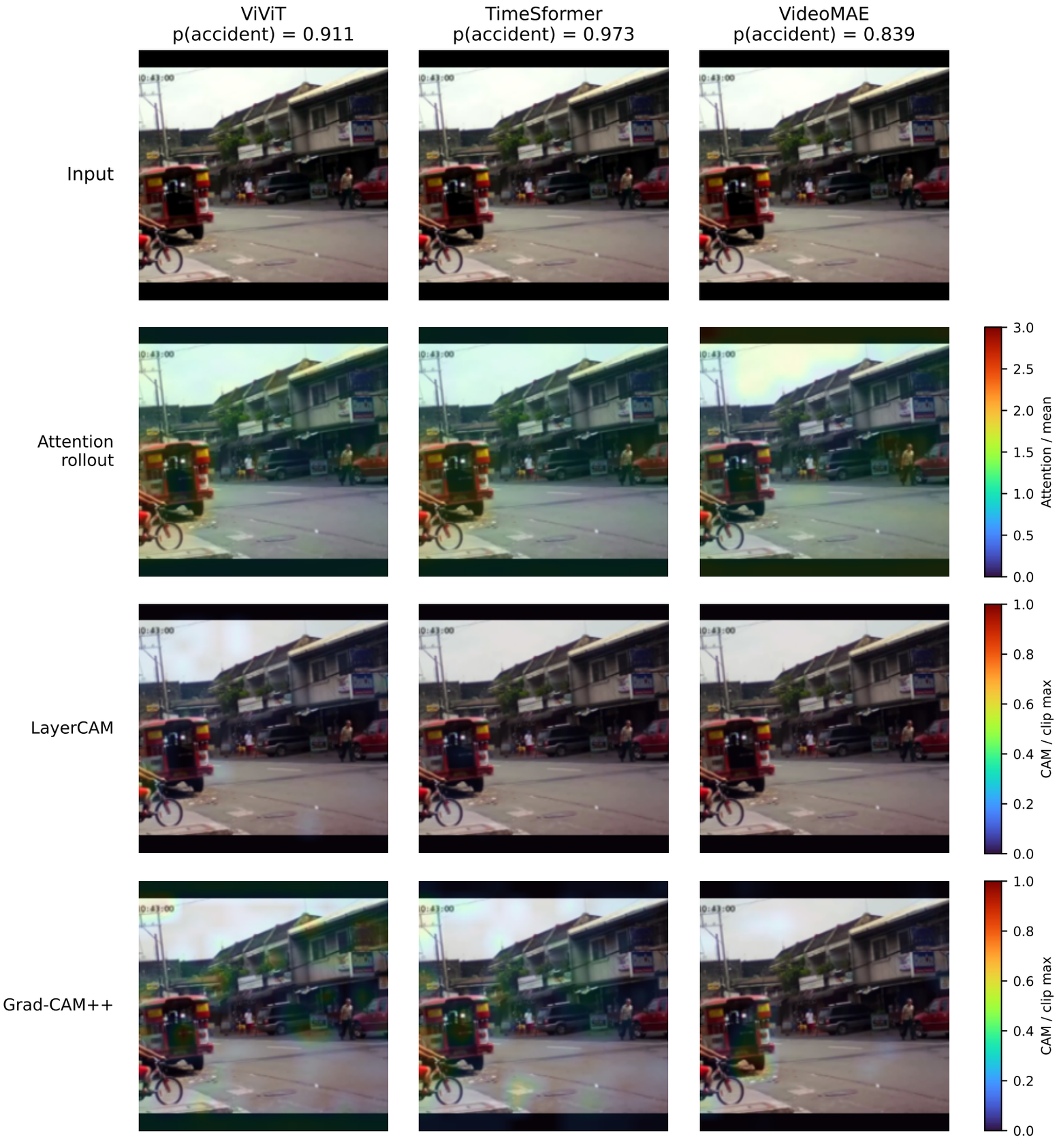
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N10 | Non-accident (dataset label) | t = 188.56 s

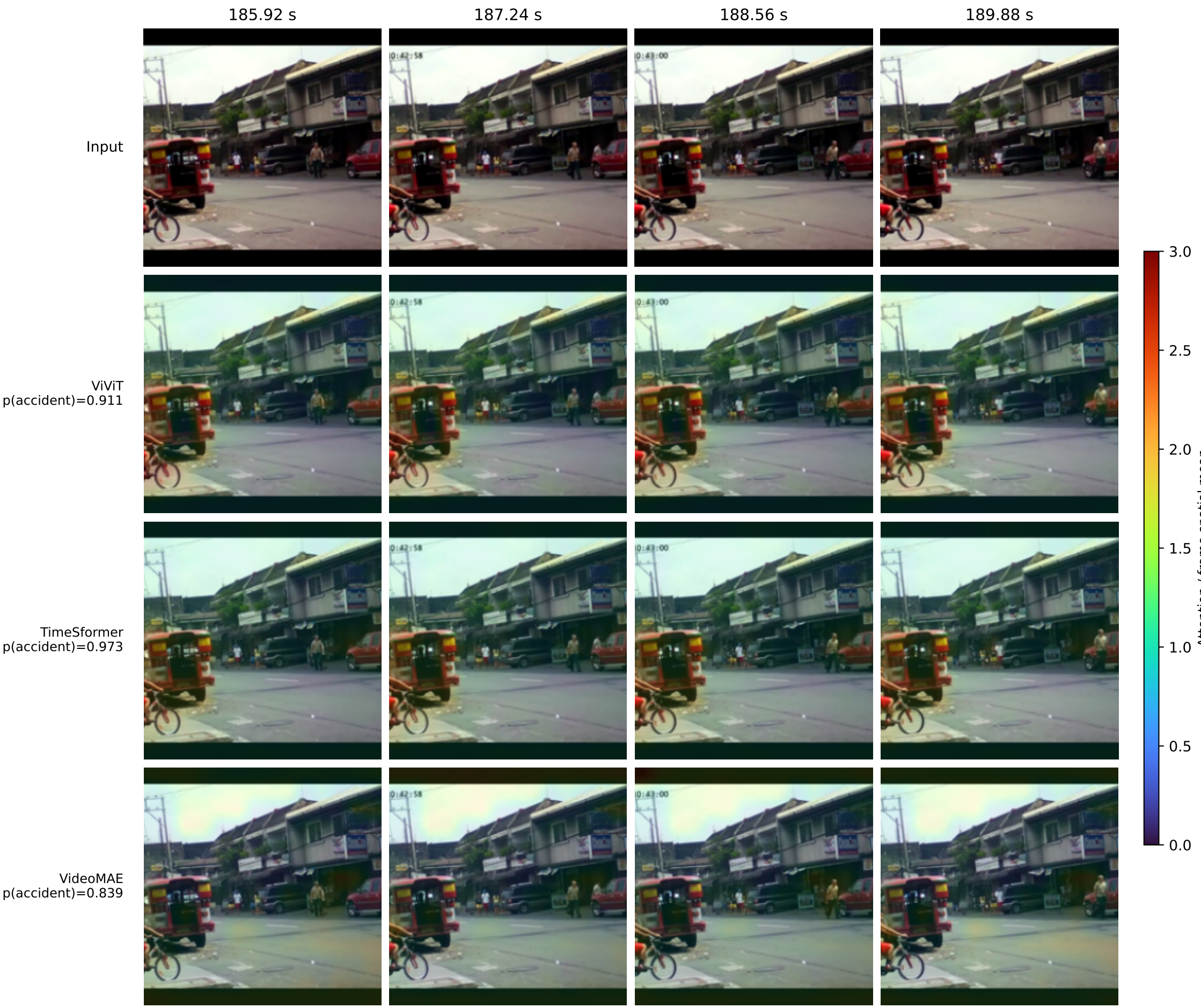
Grouped retraining, seed 42 | same displayed RGB frame | fixed center 4-second input



LayerCAM / Grad-CAM++ target: accident logit, for both labels. Rollout is not class-specific.
Maps are interpolated in time for display. Each CAM uses its own full-clip maximum.
TimeSformer rollout: spatial + temporal flow; checkpoint config 8 / input 32 frames preserved.
No onset / object-box annotations. TimeSformer requires a corrected-config retraining control.

N10 | Attention rollout | Non-accident label

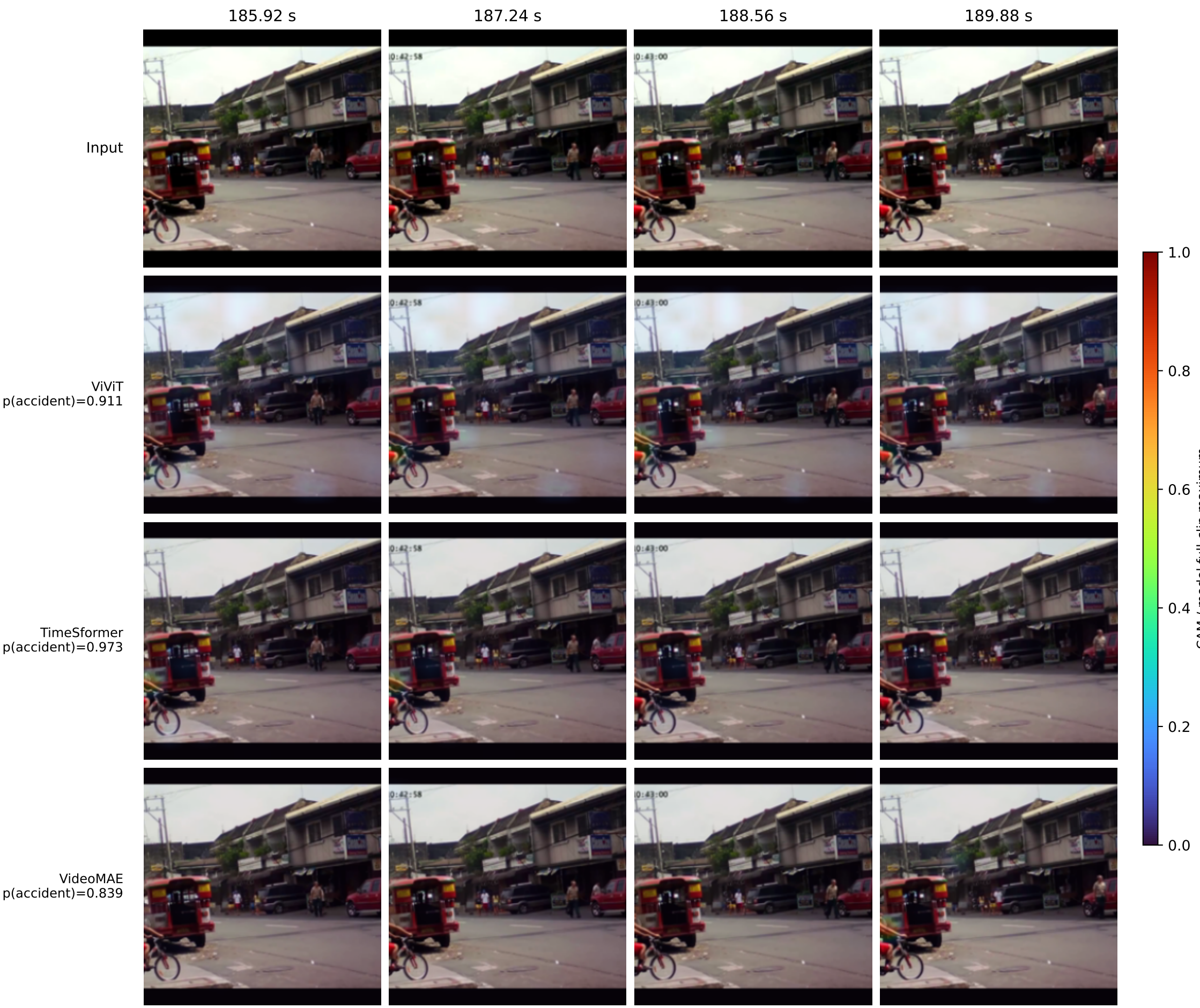
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N10 | LayerCAM | Non-accident label

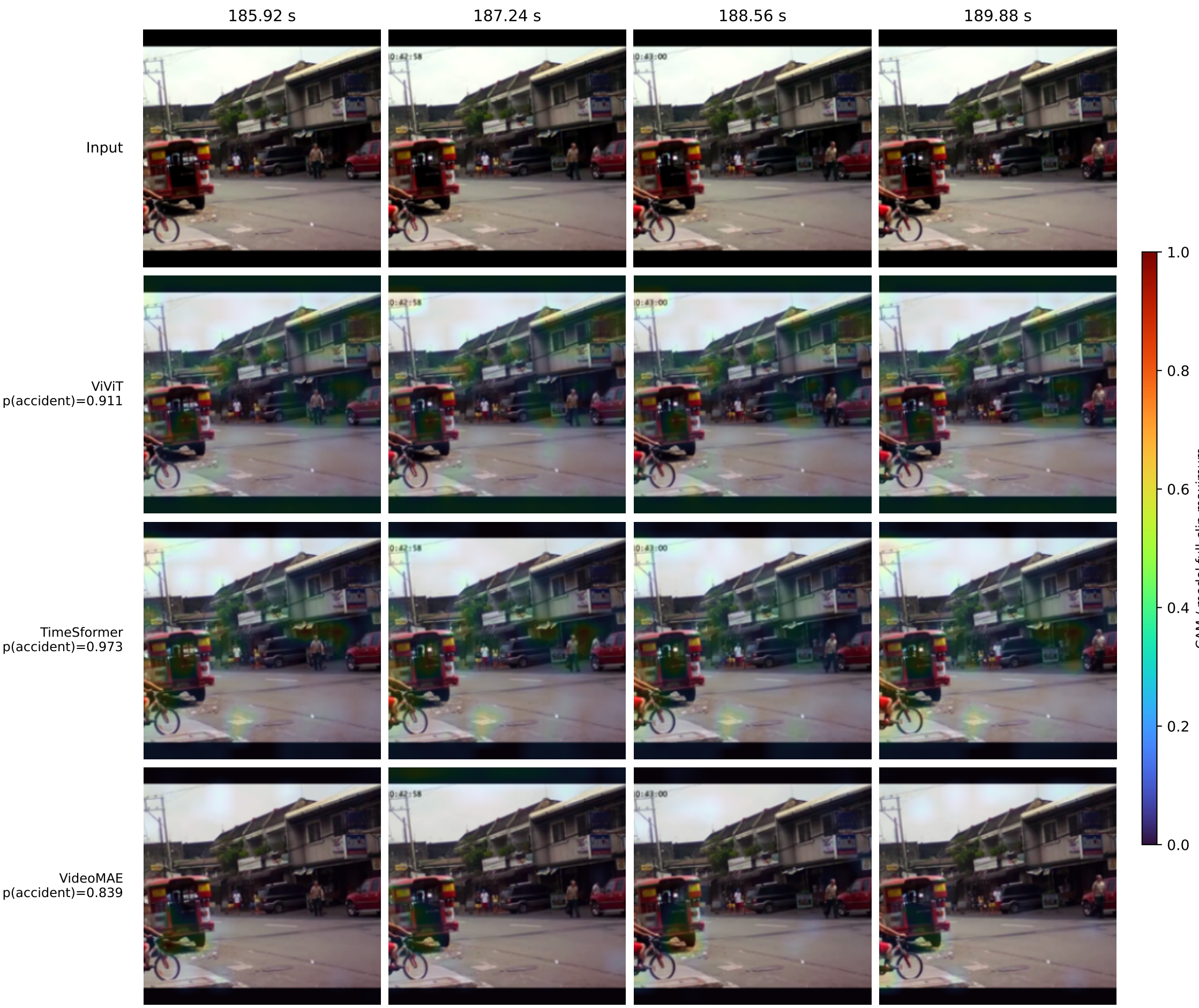
Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.

N10 | Grad-CAM++ | Non-accident label

Same four RGB display instants | model inputs: 32 / 32 / 16 frames | seed 42



Accident-logit target for CAMs, also on non-accident clips. Raw rollout is not class-specific.
Display-only interpolation. TimeSformer preserves checkpoint config 8 / input 32 frames; inverse grouping for display.
Relative colors do not establish localization accuracy or compare absolute evidence.